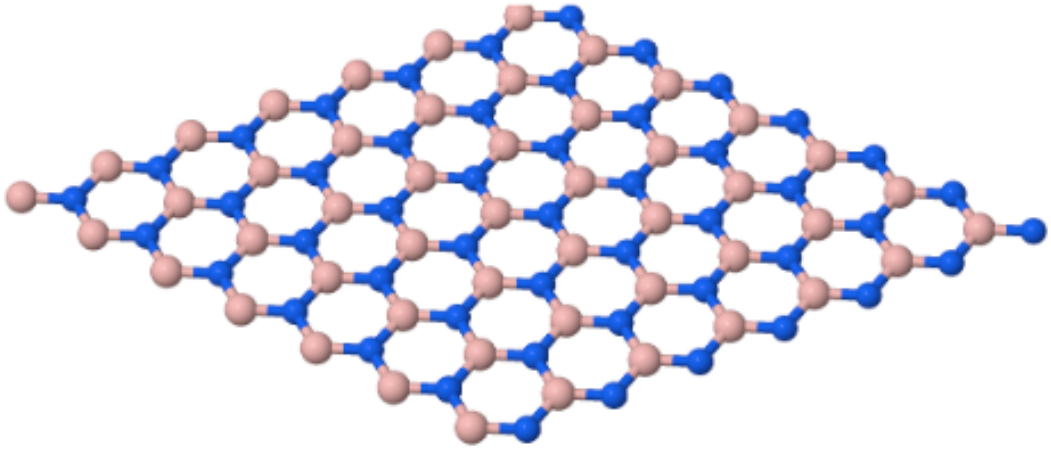
MATE70020: Modelling materials with DFT

Question Sheet 2

Problem 1: Band structure of monolayer boron nitride

Similar to graphene, hexagonal boron nitride (hBN) is a two-dimensional material with single-atom thickness. The boron and nitrogen atoms are arranged in a honeycomb lattice - see graph below. Download the input file “BN.in” and the boron and nitrogen pseudopotentials from Blackboard.



(a) In the input file the keyword K_POINTS is used to specify the grid of k-vectors of the KS wavefunctions in the first Brillouin zone. Here, a $6 \times 6 \times 1$ grid is used. This means that all $6 \times 6 \times 1 = 36$ k-vectors that are used lie in the xy-plane (which is the plane of the hBN sheet). Explain why no k-vectors which point along the z-direction are used. (1 mark)

(b) To set up the correct crystal structure of hBN, the keyword ibrav = 4 is used. Use the online documentation of Quantum Espresso (accessable via this link) to find the corresponding primitive lattice vectors. Calculate the corresponding primitive reciprocal lattice vectors. (2 marks)

(c) The first Brillouin zone of hBN is a hexagon. The K-point is located at the corner of the hexagon and the M-point is located in the middle of the side of the hexagon, see graph below. Express the position vectors of the K-point and M-point in terms of the primitive reciprocal lattice vectors. The coefficients in these expressions are the so-called crystal coordinates of the k-points. Quantum Espresso expects these coefficients as input when you use the keyword “crystal” or “crystal_b” after the keyword K_POINTS in the input file. (3 marks)

(d) Make a plot of the band structure of hBN. For this, calculate the Kohn-Sham energies along a path in the first Brillouin zone from Γ to K to M. Does hBN have a direct or an indirect band gap? What is the value of the band gap and how does it compare to the value from experiment which is 6.1 eV (Nature Communications (2019) 10:2639)? (4 marks)

